

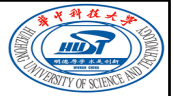
人工智能与自动化学院

模式识别与机器学习

授课教师：杨卫东 胡静



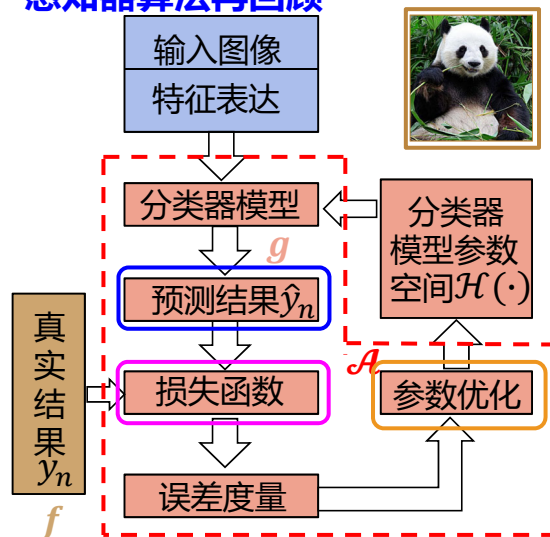
第五讲 逻辑斯蒂回归 (*Logistic Regression*)



- 5.1 逻辑斯蒂回归问题 (*Logistic Regression Problem*)
- 5.2 逻辑斯蒂回归损失 (*Logistic Regression Loss*)
- 5.3 逻辑斯蒂回归算法 (*Logistic Regression Algorithm*)
- 5.4 二元分类线性模型讨论 (*Linear Models for Binary Classification*)

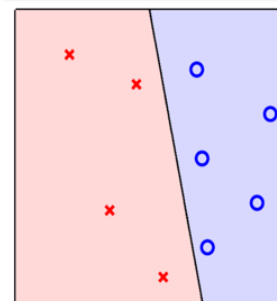
5.1 逻辑斯蒂回归问题

感知器算法再回顾



$$f(\mathbf{x}_n) = y_n \in \{1, -1\}$$

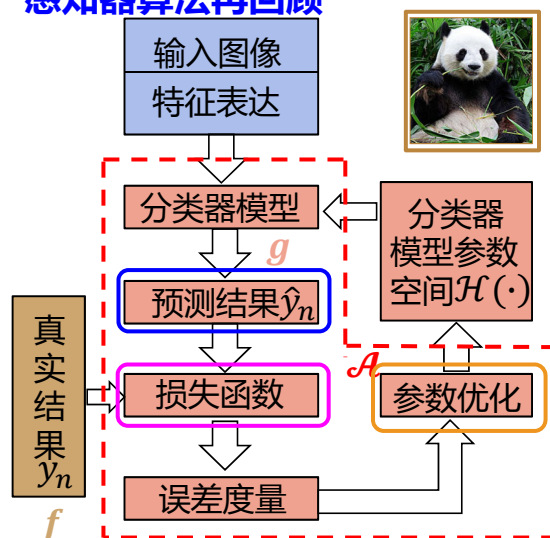
线性可分



- 设置初始分类面 (权重) $\mathbf{w}_0$
- 如果有样本分错, 就修正权重

5.1 逻辑斯蒂回归问题

感知器算法再回顾



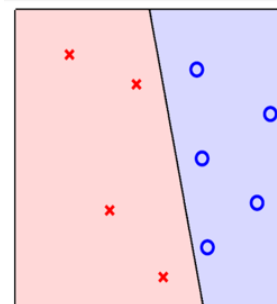
$$\hat{y}_{n(t)} = \text{sign}(\mathbf{w}_t^T \mathbf{x}_{n(t)})$$

算法收敛:

$$L_{in} = \sum_{n=1}^N \mathbb{I}[y_n \neq \hat{y}_n] = 0$$

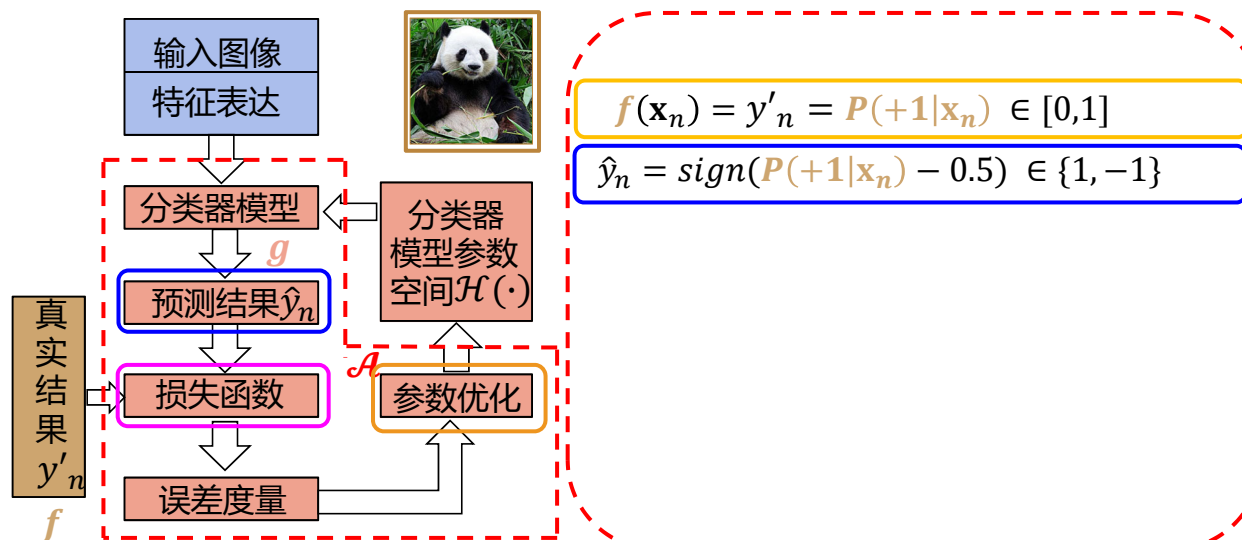
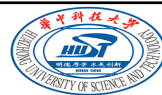
$$\mathbf{w}_{t+1} = \mathbf{w}_t + y_n \mathbf{x}_{n(t)}$$

线性可分

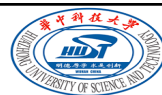


- 设置初始分类面 (权重) $\mathbf{w}_0$
- 如果有样本分错, 就修正权重

5.1 逻辑斯蒂回归问题



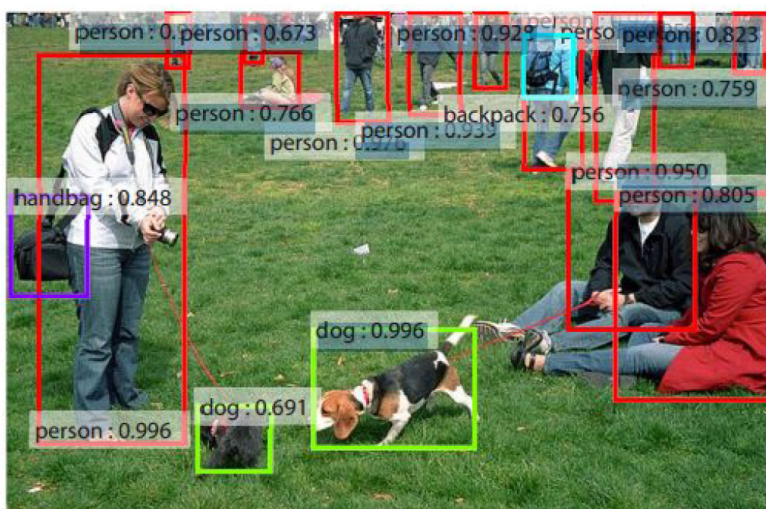
5.1 逻辑斯蒂回归问题



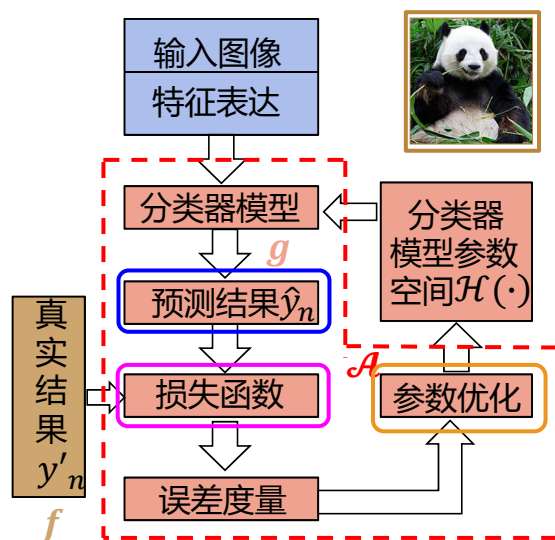
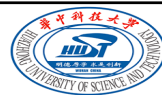
逻辑斯蒂回归应用示例

感知器算法：硬分类

逻辑斯蒂回归：软分类
(“Soft” binary classification)



5.1 逻辑斯蒂回归问题



逻辑斯蒂回归：软分类

$$f(\mathbf{x}_n) = y'_n = P(+1|\mathbf{x}_n) \in [0,1]$$

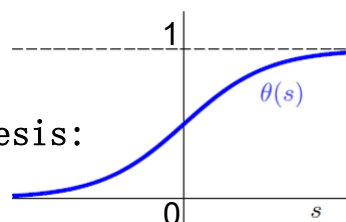
$$\hat{y}_n = \text{sign}(P(+1|\mathbf{x}_n) - 0.5) \in \{1, -1\}$$

$$\mathbf{x} = (x_1, x_2, \dots, x_d)^T$$

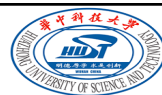
$$s = \sum_{i=1}^d w_i x_i$$

logistic hypothesis:

$$h(\mathbf{x}) = \theta(\mathbf{w}^T \mathbf{x})$$



5.1 逻辑斯蒂回归问题



逻辑斯蒂函数

$$\theta(-\infty) = 0;$$

$$\theta(0) = \frac{1}{2};$$

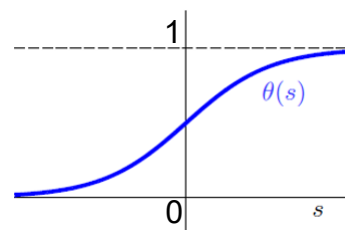
$$\theta(\infty) = 1$$

$$\theta(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

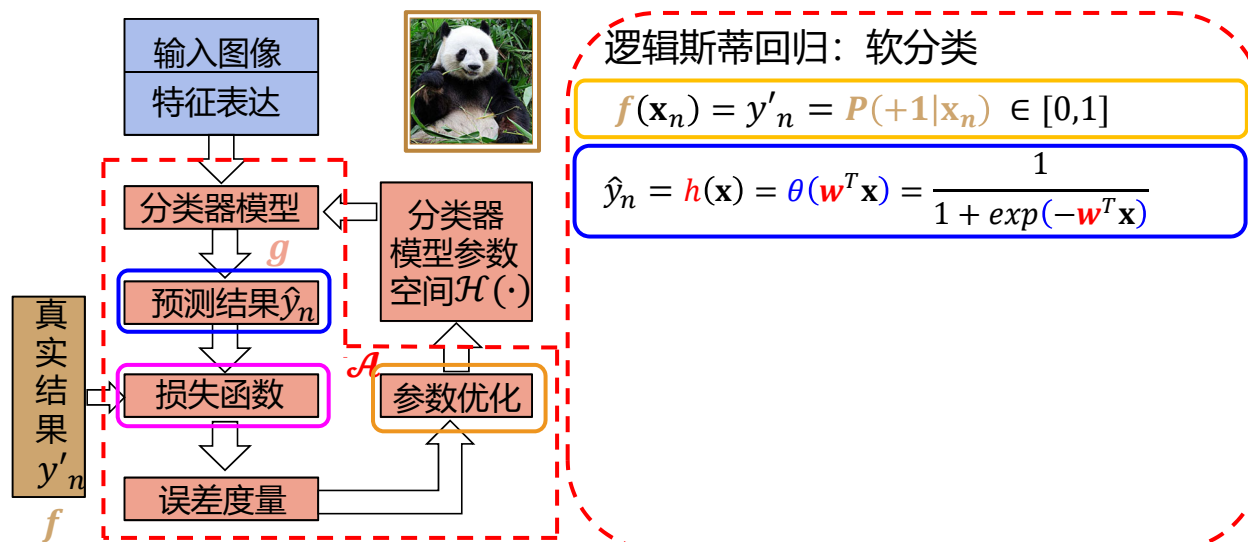
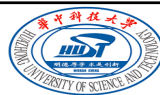
---- Sigmoid 函数：平滑 (Smooth)、单调 (Monotonic)

逻辑斯蒂回归用如下模型来估计 $f(\mathbf{x}_n)$

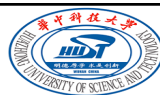
$$h(\mathbf{x}) = \theta(\mathbf{w}^T \mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^T \mathbf{x})}$$



5.1 逻辑斯蒂回归问题

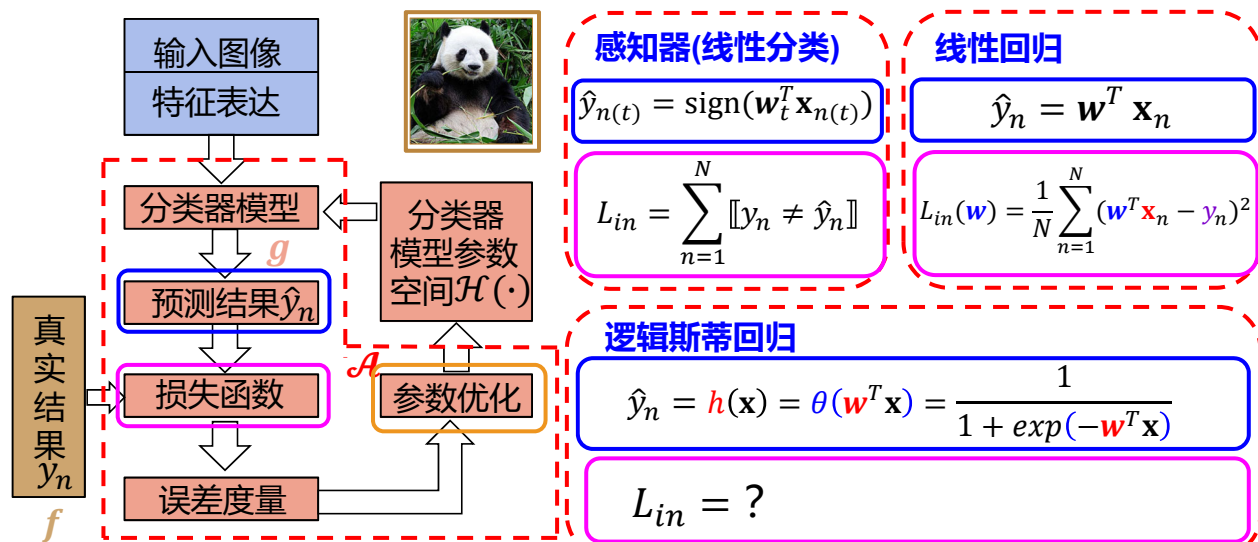
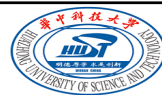


第五讲 逻辑斯蒂回归 (Logistic Regression)

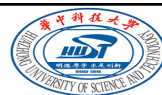


- 5.1 逻辑斯蒂回归问题 (Logistic Regression Problem)
- 5.2 逻辑斯蒂回归损失 (Logistic Regression Loss)
- 5.3 逻辑斯蒂回归算法 (Logistic Regression Algorithm)
- 5.4 二元分类线性模型讨论 (Linear Models for Binary Classification)

5.2 逻辑斯蒂回归损失



5.2 逻辑斯蒂回归损失



逻辑斯蒂回归可以使用平方误差作为损失函数吗?

$$L_{in}(\mathbf{w}) = (\mathbf{w}^T \mathbf{x} - y)^2$$

5.2 逻辑斯蒂回归损失



逻辑斯蒂回归可以使用平方误差作为损失函数吗？

$$L_{in}(\mathbf{w}) = (\theta(\mathbf{w}^T \mathbf{x}) - y')^2$$

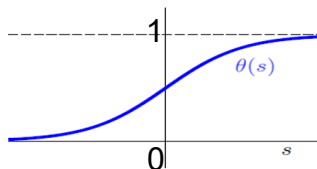
5.2 逻辑斯蒂回归损失



逻辑斯蒂回归可以使用平方误差作为损失函数吗？

$$L_{in}(\mathbf{w}) = (\theta(\mathbf{w}^T \mathbf{x}) - 1)^2$$

$$\frac{\partial L_{in}(\mathbf{w}, \mathbf{x}, y)}{\partial w_i} = 2(\theta(\mathbf{w}^T \mathbf{x}) - 1) \underbrace{\theta(\mathbf{w}^T \mathbf{x})(1 - \theta(\mathbf{w}^T \mathbf{x}))}_{\frac{\partial \theta(z)}{\partial z}} y x_i$$



$$\text{if } (y\mathbf{w}^T \mathbf{x}) > 0 \quad \forall L_{in}(\mathbf{w}, \mathbf{x}, y) = 0$$

$$\text{if } (y\mathbf{w}^T \mathbf{x}) < 0 \quad \forall L_{in}(\mathbf{w}, \mathbf{x}, y) = 0$$



5.2 逻辑斯蒂回归损失



逻辑斯蒂回归

$$f(\mathbf{x}_n) = y'_n = P(+1|\mathbf{x}_n) \in [0,1]$$

(理想)训练样本:

$$(\mathbf{x}_1, y'_1 = 0.9 = P(+1|\mathbf{x}_1))$$

$$(\mathbf{x}_2, y'_2 = 0.2 = P(+1|\mathbf{x}_2))$$

$\vdots$

$$(\mathbf{x}_N, y'_N = 0.6 = P(+1|\mathbf{x}_N))$$

实际训练样本(含噪标签):

$$(\mathbf{x}_1, y_1 = 0 = 1 \sim P(+1|\mathbf{x}_1))$$

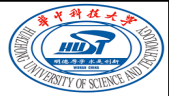
$$(\mathbf{x}_2, y_2 = \times = -1 \sim P(+1|\mathbf{x}_2))$$

$\vdots$

$$(\mathbf{x}_N, y_N = \times = -1 \sim P(+1|\mathbf{x}_N))$$

$$L_{in} = ?$$

5.2 逻辑斯蒂回归损失



逻辑斯蒂回归的最佳解:

$$g = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N -\ln \theta(y_n \mathbf{w}^T \mathbf{x}_n)$$

$$\theta(y_n \mathbf{w}^T \mathbf{x}_n) = \frac{1}{1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n)}$$

$$g = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n))$$

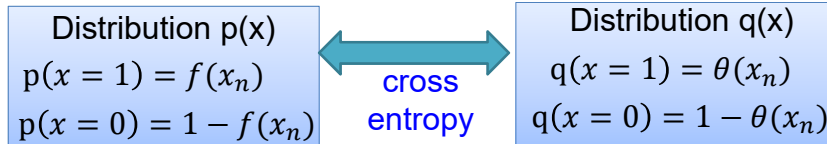
交叉熵损失
(Cross-Entropy Loss)

$$L_{in} = \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n))$$

5.2 逻辑斯蒂回归损失



交叉熵介绍:



$$s = w^T x$$

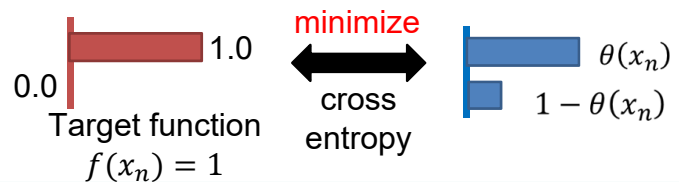
$$\theta(s) = \frac{1}{1 + \exp(-s)}$$

$$H(p, q) = - \sum_x p(x) \ln(q(x))$$

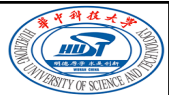
$$H(f(x_n), \theta(x_n)) = \sum_n -[f(x_n) \ln \theta(x_n) + (1 - f(x_n)) \ln (1 - \theta(x_n))]$$

$$H(f(x_n), \theta(x_n)) = \sum_n -[\ln \theta(x_n)]$$

$$H(f(x_n), \theta(x_n)) = \sum_n [\ln(1 + \exp(-ys))]$$



5.2 逻辑斯蒂回归损失



逻辑斯蒂回归的最佳解:

$$g = \underset{w}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N -\ln \theta(y_n w^T x_n)$$

$$\theta(y_n w^T x_n) = \frac{1}{1 + \exp(-y_n w^T x_n)}$$

$$g = \underset{w}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n w^T x_n))$$

交叉熵损失
(Cross-Entropy Loss)

$$L_{in} = \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n w^T x_n))$$

5.2 逻辑斯蒂回归损失

逻辑斯蒂回归的最佳解:

$$\theta(y_n \mathbf{w}^T \mathbf{x}_n) = \frac{1}{1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n)}$$

交叉熵损失
(Cross-Entropy Loss)

$$\text{令: } s = \mathbf{w}^T \mathbf{x}$$

$$\mathbf{g} = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N -\ln \theta(y_n \mathbf{w}^T \mathbf{x}_n)$$

$$\mathbf{g} = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n))$$

$$L_{in} = \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-ys))$$

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$$\text{令: } s = \mathbf{w}^T \mathbf{x}$$

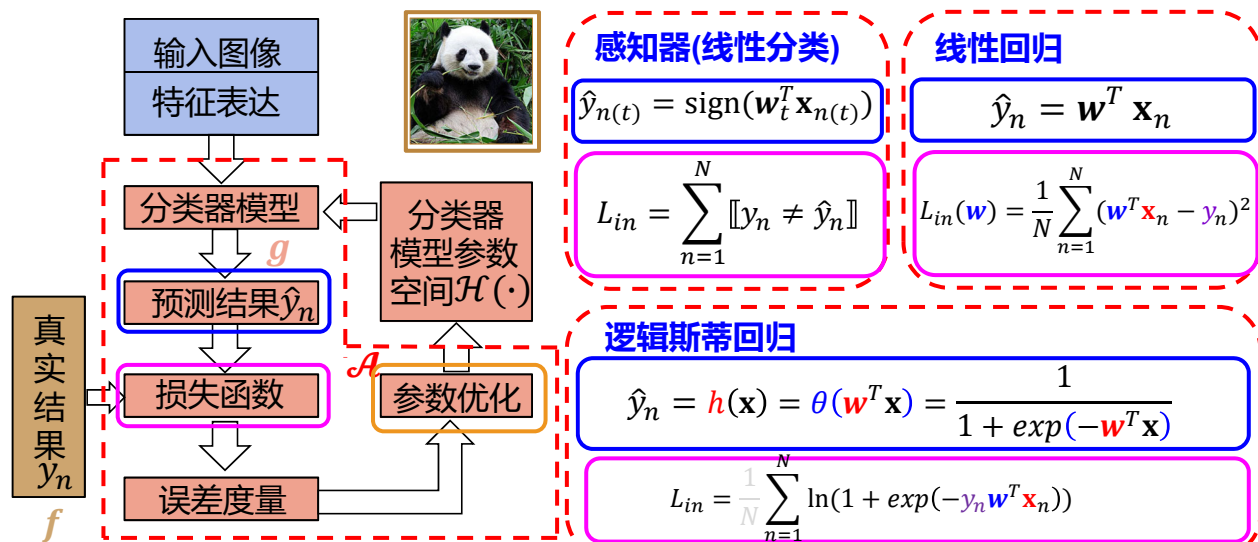
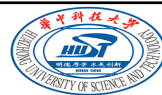
$$\mathbf{g} = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N -\ln \theta(y_n \mathbf{w}^T \mathbf{x}_n)$$

$$\mathbf{g} = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n))$$

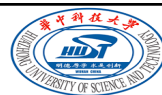
$$L_{in} = \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-ys))$$

$$H(f(x_n), \theta(x_n)) = \sum_n [\ln(1 + \exp(-ys))]$$

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$$g = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n))$$

交叉熵损失
(Cross-Entropy Loss)

$$L_{in} = \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n))$$

令: $s = \mathbf{w}^T \mathbf{x}$

$$L_{in} = \ln(1 + \exp(-ys))$$

5.2 逻辑斯蒂回归损失



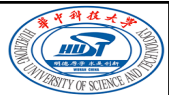
交叉熵损失梯度:

$$L_{in} = \ln(1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n))$$

$$\begin{aligned} \frac{\partial L_{in}(\mathbf{w}, \mathbf{x}, y)}{\partial w_i} &= \frac{\partial \ln(\blacksquare)}{\partial \blacksquare} \frac{\partial (1 + \exp(\bullet))}{\partial \bullet} \frac{\partial (-y \mathbf{w}^T \mathbf{x})}{\partial w_i} \\ &= \frac{1}{\blacksquare} \exp(\bullet) (-yx_i) = \frac{\exp(\bullet)}{1 + \exp(\bullet)} (-yx_i) \end{aligned}$$

$$\nabla L_{in}(\mathbf{w}, \mathbf{x}, y) = \theta(-y \mathbf{w}^T \mathbf{x}) (-y \mathbf{x})$$

5.2 逻辑斯蒂回归损失



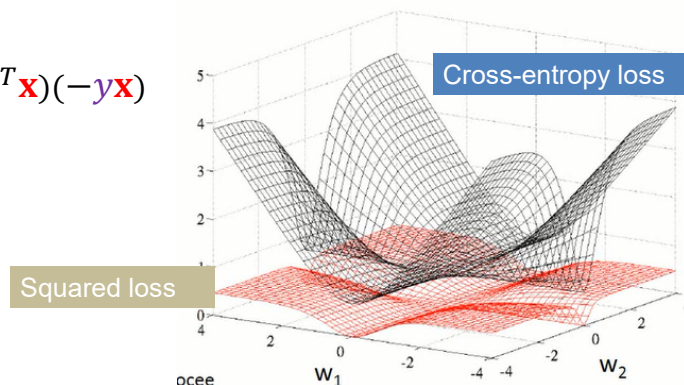
交叉熵损失与平方损失的梯度对比:

平方损失的梯度:

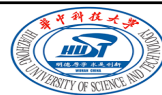
$$\nabla L_{in}(\mathbf{w}, \mathbf{x}, y) = 2(\theta(y \mathbf{w}^T \mathbf{x}) - 1) \theta(y \mathbf{w}^T \mathbf{x}) (1 - \theta(y \mathbf{w}^T \mathbf{x})) y \mathbf{x}$$

交叉熵损失的梯度:

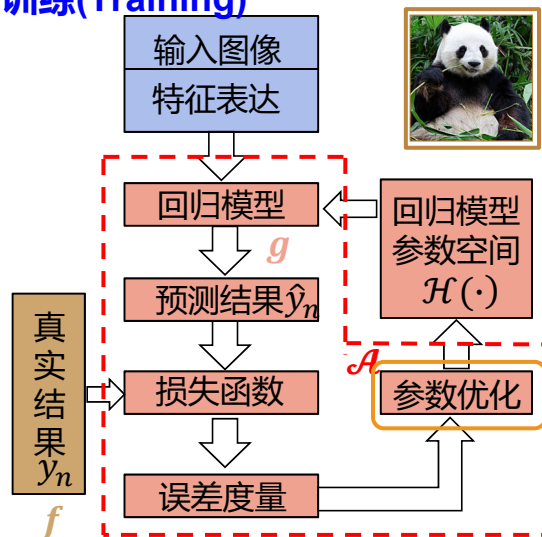
$$\nabla L_{in}(\mathbf{w}, \mathbf{x}, y) = \theta(-y \mathbf{w}^T \mathbf{x}) (-y \mathbf{x})$$



3.3 梯度下降法（引）



训练(Training)



随机梯度下降法(SGD):

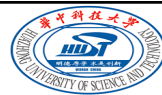
$$\nabla L_{in}(\mathbf{w}) = \sum_{n=1}^B (\mathbf{w}^T \mathbf{x}_n - y_n) \mathbf{x}_n$$

$$\mathbf{m}_{i,t+1} = \lambda \mathbf{m}_{i,t} - \frac{\eta}{\sigma_{i,t}} \frac{\partial L_{in}}{\partial \mathbf{w}_{i,t}}$$

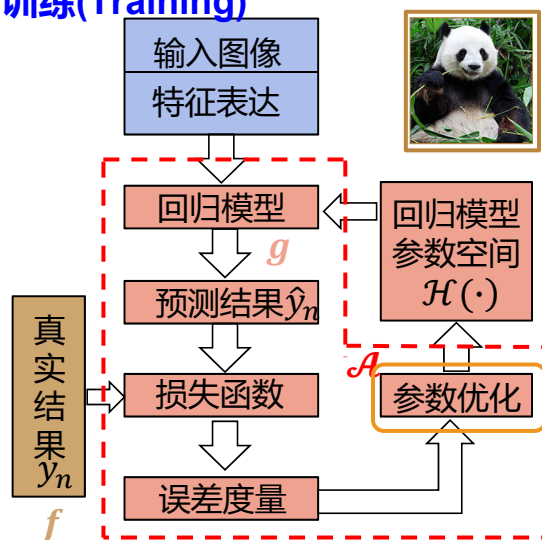
$$\mathbf{w}_{i,t+1} \leftarrow \mathbf{w}_{i,t} + \mathbf{m}_{i,t+1}$$

- 问题1: 学习率 η 如何取值?
- 问题2: 梯度为0就能得到最佳解?
- 问题3: 训练样本批量大小的影响?
- 问题4: 损失函数的影响?

3.3 梯度下降法（引）



训练(Training)



随机梯度下降法(SGD):

$$\nabla L_{in}(\mathbf{w}) = \sum_{n=1}^B (\mathbf{w}^T \mathbf{x}_n - y_n) \mathbf{x}_n$$

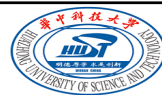
$$\mathbf{m}_{i,t+1} = \lambda \mathbf{m}_{i,t}$$

$$\mathbf{w}_{i,t+1} \leftarrow \mathbf{w}_{i,t}$$

不同损失函数
在梯度下降时
的速度不同

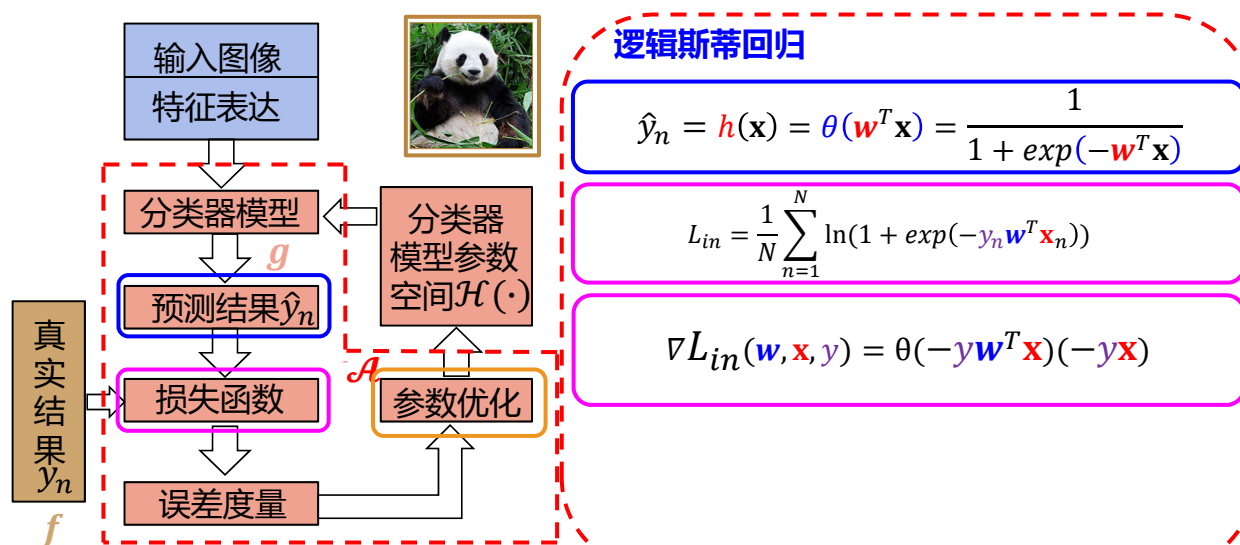
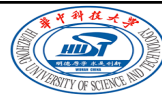
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第五讲 逻辑斯蒂回归 (Logistic Regression)

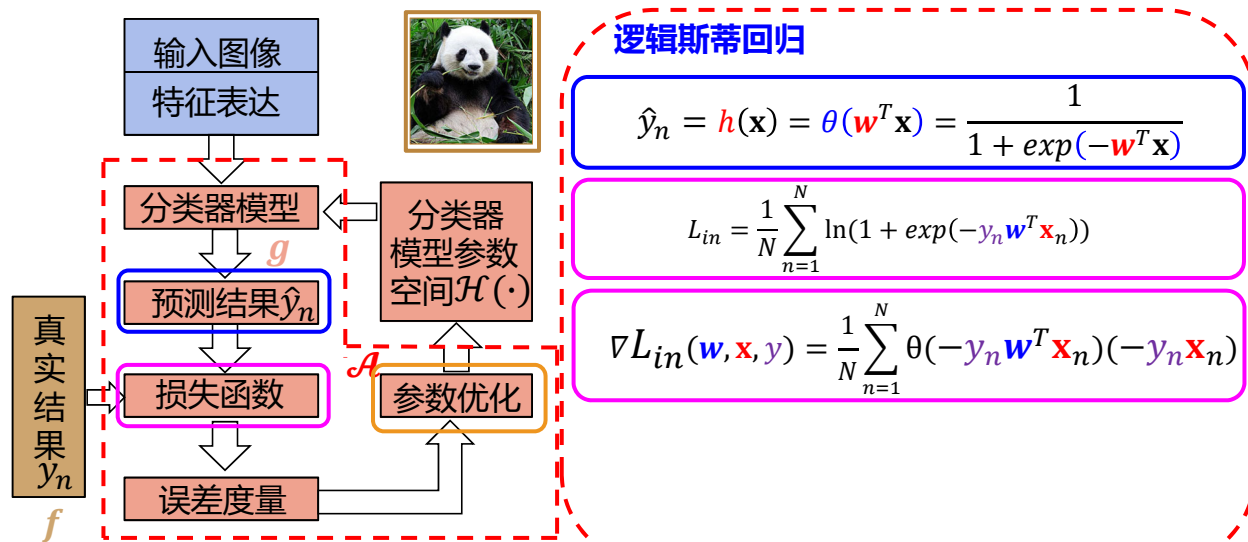
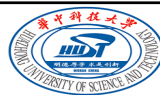


- 5.1 逻辑斯蒂回归问题 (Logistic Regression Problem)
- 5.2 逻辑斯蒂回归损失 (Logistic Regression Loss)
- 5.3 逻辑斯蒂回归算法 (Logistic Regression Algorithm)
- 5.4 二元分类线性模型讨论 (Linear Models for Binary Classification)

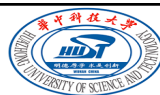
5.2 逻辑斯蒂回归损失



5.2 逻辑斯蒂回归损失



3.2 线性回归算法



梯度下降法实现逻辑斯蒂回归

- 初始化权向量 $\mathbf{w}_0$
- **for** $t = 0, 1, 2, \dots$ (t 代表迭代次数)
 - ① 计算梯度: $\nabla L_{in}(\mathbf{w}_t) = \frac{1}{N} \sum_{n=1}^N \theta(-y_n \mathbf{w}_t^T \mathbf{x}_n) (-y_n \mathbf{x}_n)$
 - ② 对权向量 $\mathbf{w}_t$ 进行更新: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta \nabla L_{in}(\mathbf{w}_t)$

...直到 $\nabla L_{in}(\mathbf{w}) = \mathbf{0}$, 或者迭代足够多次数

返回最终的 $\mathbf{w}_{t+1}$ 作为学到的 g

3.2 线性回归算法



梯度下降法实现逻辑斯蒂回归

- 初始化权向量 w_0 **Stochastic Gradient Descent(SGD)**

- **for** $t = 0, 1, 2, \dots$ (t 代表迭代次数)

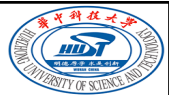
① 计算梯度: $\nabla L_{in}(w_t) = \frac{1}{B} \sum_{n=1}^B \theta(-y_n w_t^T x_n) (-y_n x_n)$

② 对权向量 w_t 进行更新: $w_{t+1} \leftarrow w_t - \eta \nabla L_{in}(w_t)$

...直到 $\nabla L_{in}(w) = 0$, 或者迭代足够多次数

返回最终的 w_{t+1} 作为学到的 g

第五讲 逻辑斯蒂回归 (Logistic Regression)



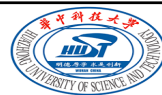
5.1 逻辑斯蒂回归问题 (Logistic Regression Problem)

5.2 逻辑斯蒂回归损失 (Logistic Regression Loss)

5.3 逻辑斯蒂回归算法 (Logistic Regression Algorithm)

5.4 二元分类线性模型讨论 (Linear Models for Binary Classification)

5.4 二元分类线性模型讨论



三种线性模型比较

线性分类(感知器):

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x})$$

$$L_{in} = \sum_{n=1}^N \mathbb{I}[y_n \neq \hat{y}_n]$$

NP难问题

线性回归:

$$h(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$$

$$L_{in} = \frac{1}{N} \sum_{n=1}^N (\mathbf{w}^T \mathbf{x}_n - y_n)^2$$

凸函数、易优化、解析解

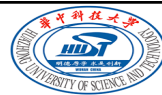
逻辑斯蒂回归:

$$h(\mathbf{x}) = \theta(\mathbf{w}^T \mathbf{x})$$

$$L_{in} = \frac{1}{N} \sum_{n=1}^N \ln(1 + \exp(-y_n \mathbf{w}^T \mathbf{x}_n))$$

凸函数、平滑、梯度下降

5.4 二元分类线性模型讨论



三种线性模型用于二元分类时(即: $y \in \{+1, -1\}$) 损失函数比较

样本特征向量 $\mathbf{x}$ 与模型的权向量 $\mathbf{w}$ 的内积用 s 表示: $s = \mathbf{w}^T \mathbf{x}$

线性分类(感知器):

$$h(\mathbf{x}) = \text{sign}(s)$$

$$L_{in} = \mathbb{I}[h(\mathbf{x}) \neq y]$$

$$L_{0/1}(s, y) = \mathbb{I}[\text{sign}(s) \neq y]$$

$$= \mathbb{I}[\text{sign}(ys) \neq 1]$$

线性回归:

$$h(\mathbf{x}) = s$$

$$L_{in} = (h(\mathbf{x}) - y)^2$$

$$L_{sq}(s, y) = (s - y)^2$$

$$= (ys - 1)^2$$

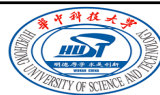
逻辑斯蒂回归:

$$h(\mathbf{x}) = \theta(s)$$

$$L_{in} = -\ln(h(y\mathbf{x}))$$

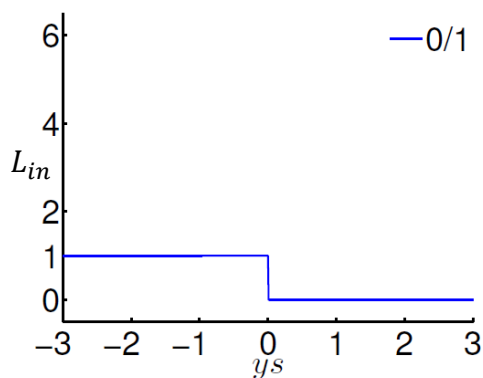
$$L_{ce} = \ln(1 + \exp(-ys))$$

5.4 二元分类线性模型讨论



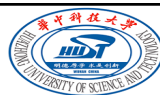
三种线性模型用于二元分类时(即: $y \in \{+1, -1\}$) 损失函数比较

样本特征向量 $\mathbf{x}$ 与模型的权向量 $\mathbf{w}$ 的内积用 s 表示: $s = \mathbf{w}^T \mathbf{x}$



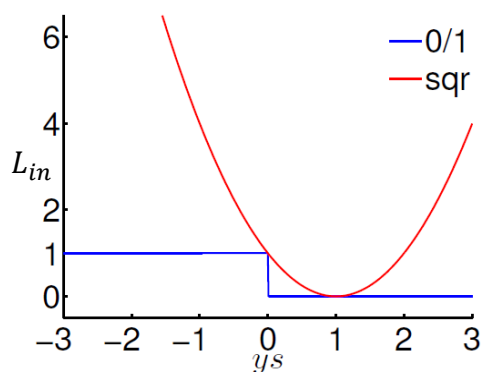
$$0/1 \quad L_{0/1}(s, y) = \mathbb{I}[\text{sign}(ys) \neq 1]$$

5.4 二元分类线性模型讨论



三种线性模型用于二元分类时(即: $y \in \{+1, -1\}$) 损失函数比较

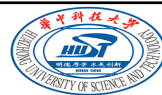
样本特征向量 $\mathbf{x}$ 与模型的权向量 $\mathbf{w}$ 的内积用 s 表示: $s = \mathbf{w}^T \mathbf{x}$



$$0/1 \quad L_{0/1}(s, y) = \mathbb{I}[\text{sign}(ys) \neq 1]$$

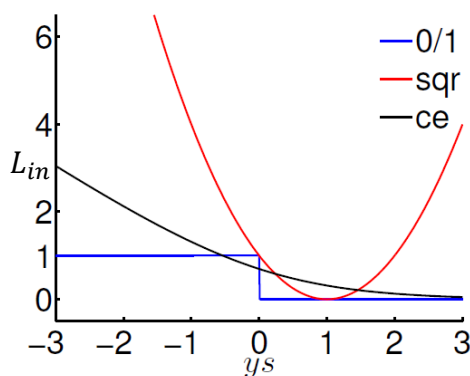
$$\text{sqr} \quad L_{sqr}(s, y) = (ys - 1)^2$$

5.4 二元分类线性模型讨论



三种线性模型用于二元分类时(即: $y \in \{+1, -1\}$)损失函数比较

样本特征向量 $\mathbf{x}$ 与模型的权向量 $\mathbf{w}$ 的内积用 s 表示: $s = \mathbf{w}^T \mathbf{x}$

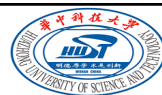


0/1 $L_{0/1}(s, y) = \mathbb{I}[\text{sign}(ys) \neq 1]$

sqr $L_{\text{sqr}}(s, y) = (ys - 1)^2$

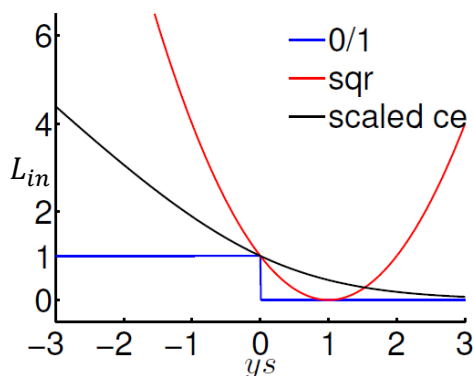
ce $L_{\text{ce}}(s, y) = \ln(1 + \exp(-ys))$

5.4 二元分类线性模型讨论



三种线性模型用于二元分类时(即: $y \in \{+1, -1\}$)损失函数比较

样本特征向量 $\mathbf{x}$ 与模型的权向量 $\mathbf{w}$ 的内积用 s 表示: $s = \mathbf{w}^T \mathbf{x}$



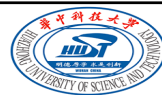
0/1 $L_{0/1}(s, y) = \mathbb{I}[\text{sign}(ys) \neq 1]$

sqr $L_{\text{sqr}}(s, y) = (ys - 1)^2$

ce $L_{\text{ce}}(s, y) = \ln(1 + \exp(-ys))$

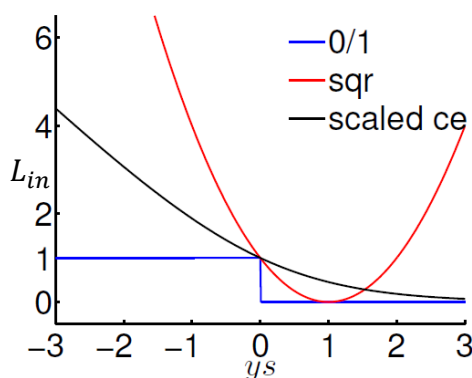
Scaled ce $L_{\text{sce}}(s, y) = \log_2(1 + \exp(-ys))$

5.4 二元分类线性模型讨论



三种线性模型用于二元分类时(即: $y \in \{+1, -1\}$) 损失函数比较

样本特征向量 $\mathbf{x}$ 与模型的权向量 $\mathbf{w}$ 的内积用 s 表示: $s = \mathbf{w}^T \mathbf{x}$



$$L_{0/1}(s, y) \leq L_{sqr}(s, y)$$

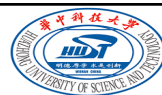
$$L_{0/1}(s, y) \leq L_{sce}(s, y)$$

$$L_{0/1}(s, y) \leq L_{ce}(s, y)$$

训练或测试时, 只要做到 $L_{sqr}(s, y)$ 或者 $L_{ce}(s, y)$ 很小, $L_{0/1}(s, y)$ 也会很小

线性回归与逻辑斯蒂回归可用于线性分类

5.4 二元分类线性模型讨论



① 在标签为 $\{+1, -1\}$ 的训练样本集 $\mathcal{D}$ 上运行线性回归/逻辑斯蒂回归算法, 得到 $\mathbf{w}^*$

② 返回分类结果: $g(\mathbf{x}) = \text{sign}(\mathbf{w}^{*T} \mathbf{x})$

线性分类(感知器):

优点: 样本线性可分时, 算法收敛有理论保障

不足: 样本非线性可分时 NP 难问题, 可用 Pocket 算法实现

线性回归:

优点: 凸函数, 最容易优化, 有解析解

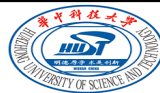
不足: 当 $|y_s|$ 很大时, $L_{0/1}(s, y)$ 的上界过于宽松

逻辑斯蒂回归:

优点: 凸函数, 易于优化

不足: 当 $y_s \ll 0$ 时, $L_{0/1}(s, y)$ 的上界过于宽松

第五讲 逻辑斯蒂回归 (Logistic Regression)



5.1 逻辑斯蒂回归问题

模型的理论输出为概率值，分类面假设空间模型用Sigmoid函数

5.2 逻辑斯蒂回归损失

用交叉熵(cross-entropy)作为损失函数

5.3 逻辑斯蒂回归算法

用梯度下降法迭代实现参数更新

5.4 二元分类线性模型讨论

三个线性模型的特点及用途